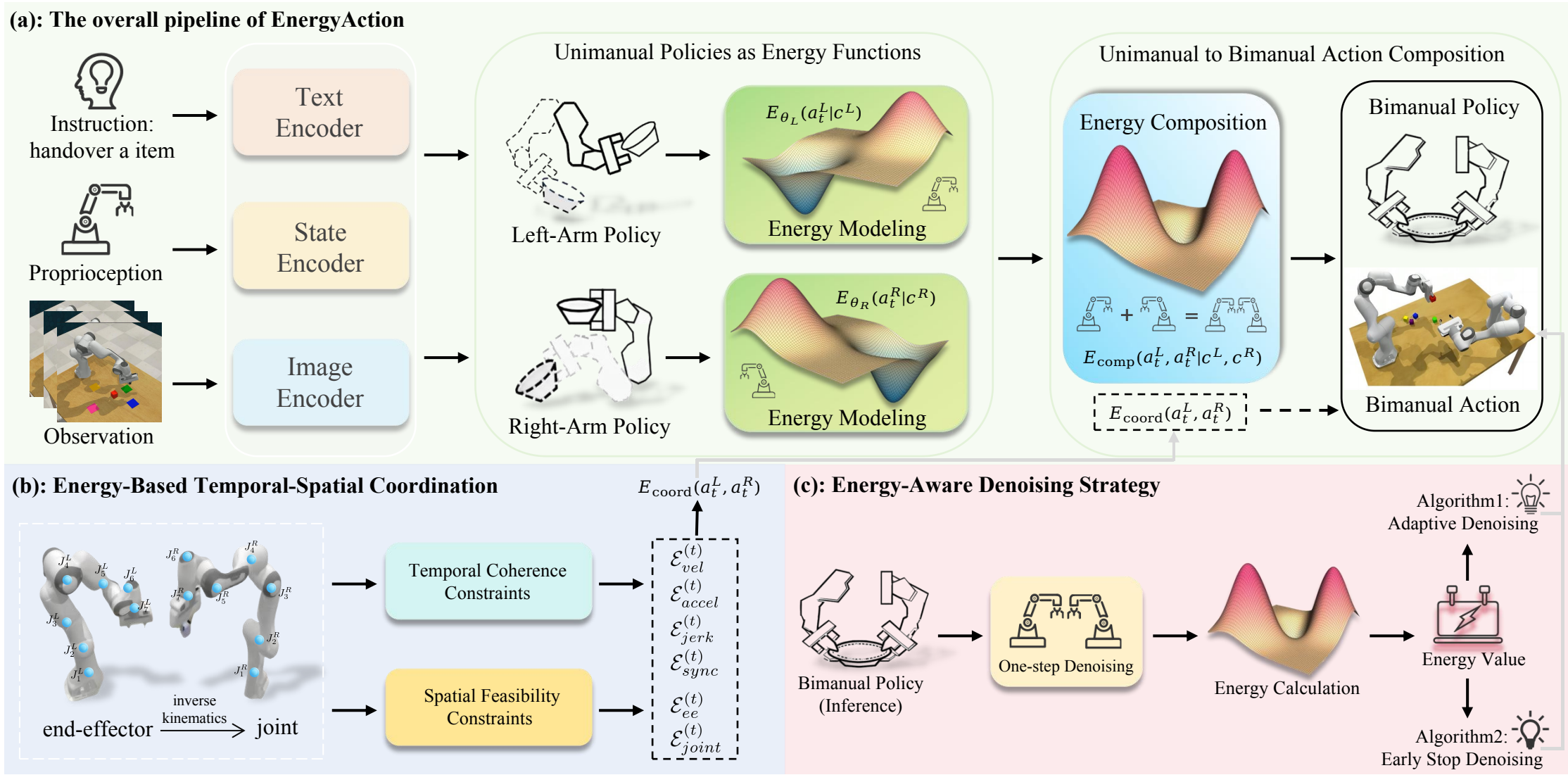
(a): The overall pipeline of EnergyAction
Instruction: handover a item
Proprioception
Observation
Text Encoder
State Encoder
Image Encoder
Unimanual Policies as Energy Functions
Left-Arm Policy
Right-Arm Policy
$E_{\theta_L}(a_t^L|c^L)$
Energy Modeling
$E_{\theta_R}(a_t^R|c^R)$
Energy Modeling
Unimanual to Bimanual Action Composition
Energy Composition
$E_{\text{comp}}(a_t^L, a_t^R|c^L, c^R)$
$E_{\text{coord}}(a_t^L, a_t^R)$
Bimanual Policy
Bimanual Action
(b): Energy-Based Temporal-Spatial Coordination
$E_{\text{coord}}(a_t^L, a_t^R)$
inverse kinematics
end-effector
joint
Temporal Coherence Constraints
Spatial Feasibility Constraints
$\mathcal{E}_{vel}^{(t)}$
$\mathcal{E}_{accel}^{(t)}$
$\mathcal{E}_{jerk}^{(t)}$
$\mathcal{E}_{sync}^{(t)}$
$\mathcal{E}_{ee}^{(t)}$
$\mathcal{E}_{joint}^{(t)}$
(c): Energy-Aware Denoising Strategy
Bimanual Policy (Inference)
One-step Denoising
Energy Calculation
Energy Value
Algorithm1: Adaptive Denoising
Algorithm2: Early Stop Denoising